

Trust-free Personalized Decentralized Learning

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- 1. Introduction
 - 2. Methodology
 - 3. Experiments
 - 4. Conclusion

Fed 1.0: Centralized FL (CFed)

Concept: The server acts as the central aggregator and content producer. Clients are merely consumers and data contributors. The server collects diverse local updates (opinions) to generate a single mainstream global model.

Advantages

- Privacy Protection
- Data Parallelism
- Democratization

Disadvantages

- Server Trust Issues
- Single Point of Failure
- Model Homogeneity

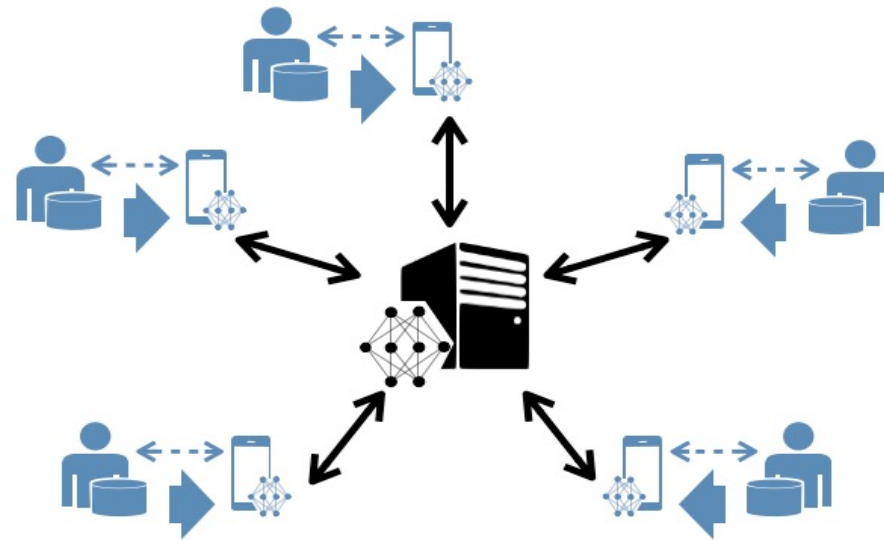


Figure 1(a). Centralized FL.

Fed 2.0: Semi-decentralized FL (SFed)

Concept: Clients act as both producers and consumers. The server functions as a "Coordinator" or "Broker" rather than a strict aggregator. It manages topology and pushes relevant content based on client profiles, allowing non-mainstream opinions to propagate within specific groups.

Advantages

- Privacy Protection
- Data Parallelism
- Personalization

Disadvantages

- Coordinator Trust Issues
- Reliability Risks
- Lack of Peer Verification

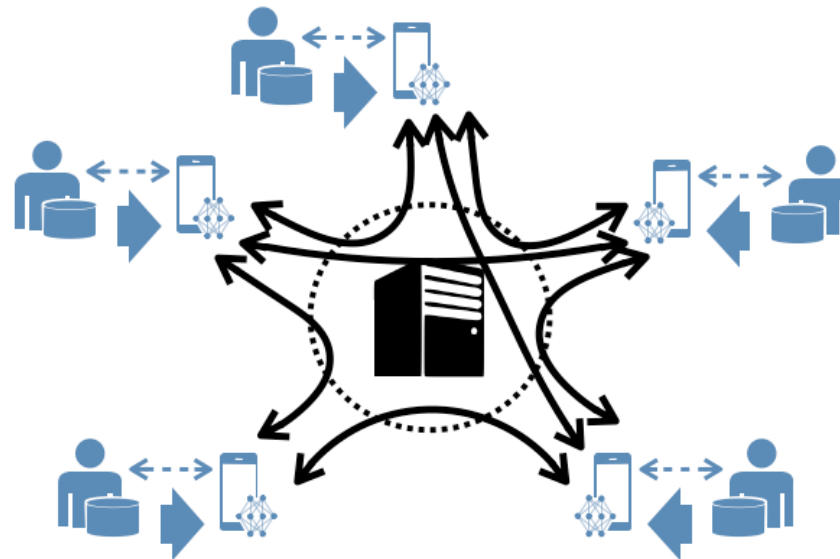


Figure 1(b). Semi-decentralized FL.

Fed 2.5: Gossip-based Decentralized FL (DFed-G)

Concept: Clients communicate directly with peers via local Gossip Protocols or physical proximity (e.g., D2D). The topology is static or random, with no central server or coordinator involved.

Advantages

Removal of Single Point of Failure
Communication Efficiency in Local Areas

Disadvantages

Limited Global View
Inefficient Neighbor Selection
Vulnerability (Lack of Trust)

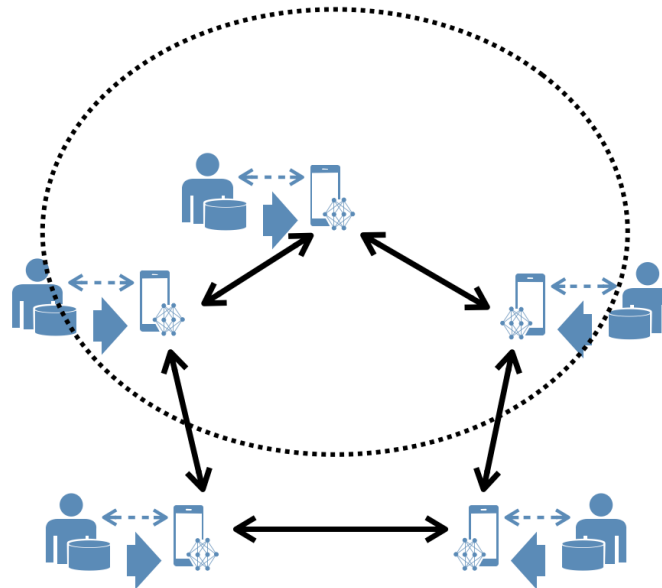


Figure 1(c). Gossip-based Decentralized FL.

What's the Gap Between Fed 2.5 to Fed 3.0?

Fed 3.0: Web-based P2P Decentralized FL (DFed-W)

Concept: Clients utilize a Public Bulletin Board (Blockchain) as a trust-free medium to discover global neighbors. Communication is autonomous and "Web-based" (logical P2P), allowing connections based on Semantic Similarity (LSH) and Verifiable Quality, rather than just physical proximity.

Advantages

- Global Visibility
- Trust-free Collaboration
- Robust Personalization

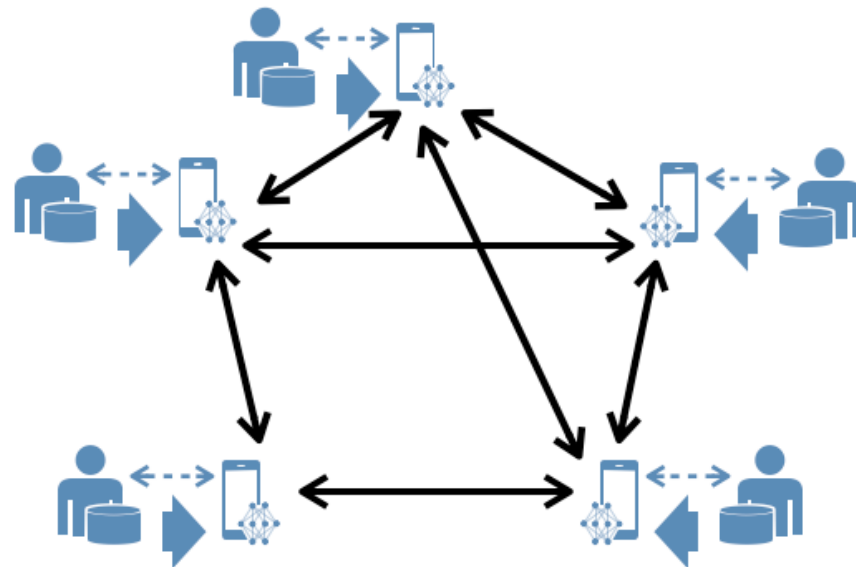


Figure 1(d). Web-based Decentralized FL.

Fed 3.0: Web-based P2P Decentralized FL (DFed-W)

Table 1. Approach Functionality Comparison

Characteristics	Silo	Fed 1.0 CFed	Fed 2.0 SFed	Fed 2.5 DFed-G	Fed 3.0 DFed-W
Communication content generated by clients	—	x	✓	✓	✓
Self-determined neighbor selection	—	—	x	✓	✓
Global perspective for neighbor selection	—	—	✓	x	✓
Quality checks in communication	—	✓	✓	x	✓
Personalized model updates	—	x	✓	✓	✓
Adversarial resilience	✓	x	x	x	✓
Trust-free communication	—	x	x	x	✓

“x” : not supported, “✓” : supported, “—” : not applicable.

Why Fed 3.0 is hard?

Challenge 1: P2P Similarity Evaluation

Why Similarity Evaluation is important?

Heterogeneity: Client models differ significantly due to training progress, non-IID data distributions, and network topology.

The Risk: Collaborating with semantically dissimilar clients degrades model convergence and performance.

Evolution of Solutions

Fed 1.0: Non-issue (Single global model enforces uniformity).

Fed 2.0: Trusted Coordinator selects neighbors based on global profiles.

Fed 2.5: Clients evaluate inter-client similarity locally (within neighborhood).

Fed 3.0: P2P similarity Evaluation in a global perspective?

Dilemma of Fed 3.0

Broadcast Model/Data? → Privacy Leakage.

Use Public Reference Data? → Masquerade Attacks (e.g., LSH Forgery).

Pairwise Verification? → Engineering Bottleneck (High Latency/Cost).

Challenge 2: P2P Quality Assessment

Why Quality Assessment is important?

Quality Variance: Communication content quality depends on local dataset size, label quality, and training rounds.

The Risk: Low-quality communication content will significantly degrade the model performance of single-hop neighbors.

Evolution of Solutions

Fed 1.0: Non-issue (Global model implies no peer variation).

Fed 2.0: Trusted Coordinator reviews content on behalf of clients.

Fed 2.5: Clients review neighbors' communication content locally (within neighborhood)..

Fed 3.0: P2P Quality Assessment in a global perspective?

Dilemma of Fed 3.0

Direct Inspection? → Privacy Leakage.

Reference Performance Check? → Overfitting (Adversaries memorize reference data).

Reputation System? → Collusion / Sybil Attacks (Hard to maintain distributed trust).

1. Introduction

▶ 2. Methodology

3. Experiments

4. Conclusion

**The story starts from the 'logits' in
knowledge distillation.**

A Quick Look in Knowledge Distillation (KG)

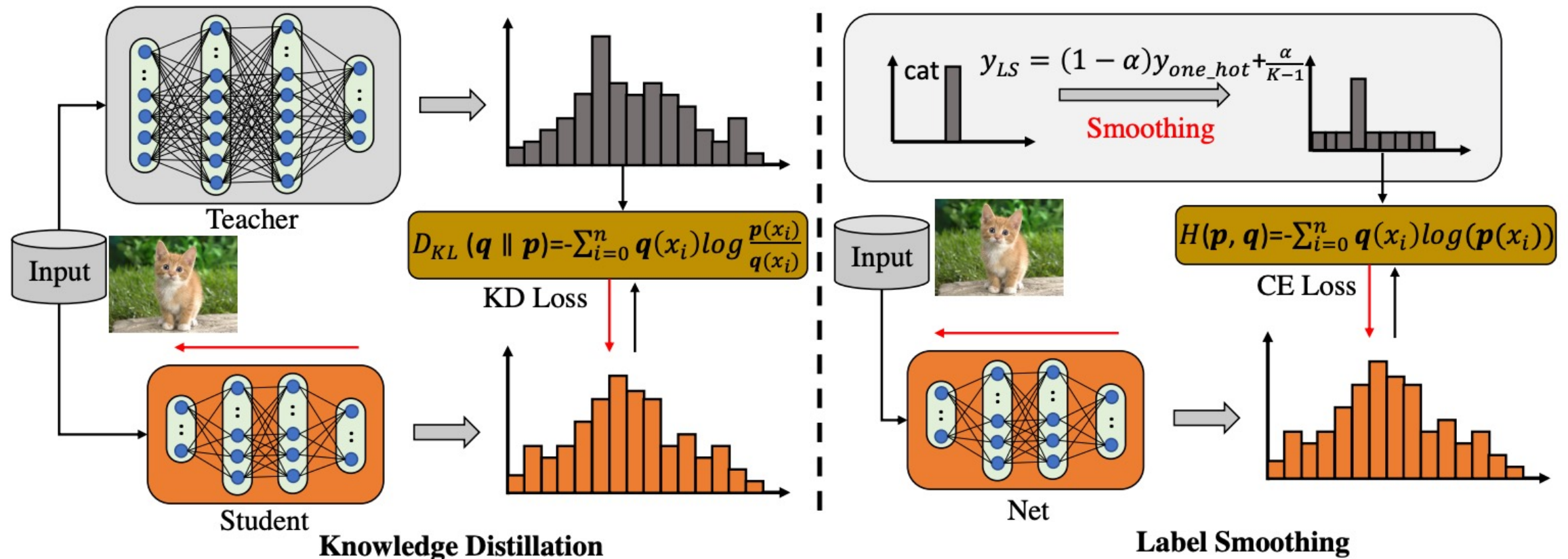


Figure 2*. Different between logits and labels.

*From Shen, Zhiqiang, et al. "Is Label Smoothing Truly Incompatible with Knowledge Distillation: An Empirical Study." *ICLR 2021*

Logits-based P2P Knowledge Sharing

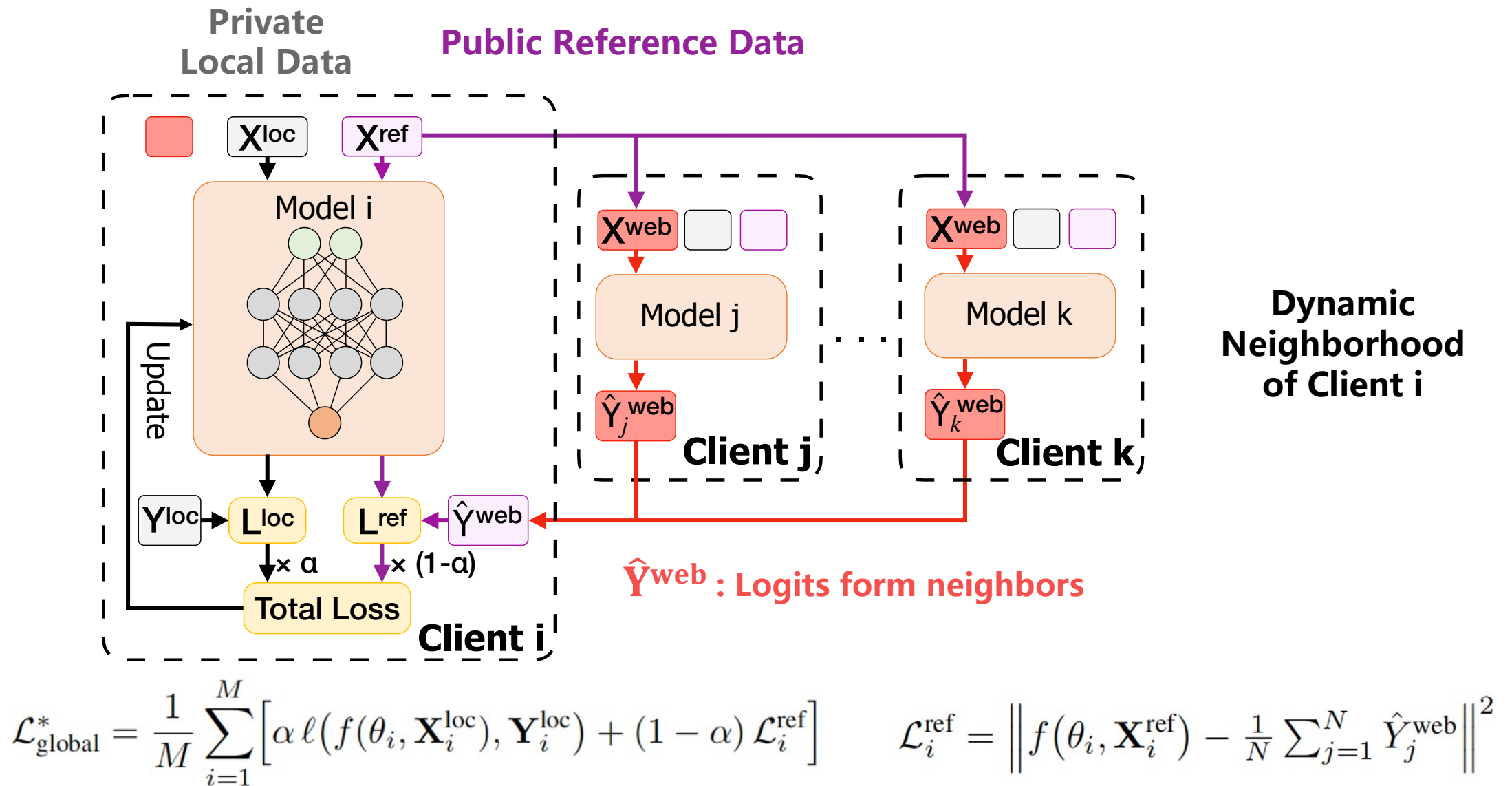


Figure 3. Client learns from neighbors via collecting logits.

What can logits do beyond knowledge sharing?

Logits-based P2P Similarity Evaluation

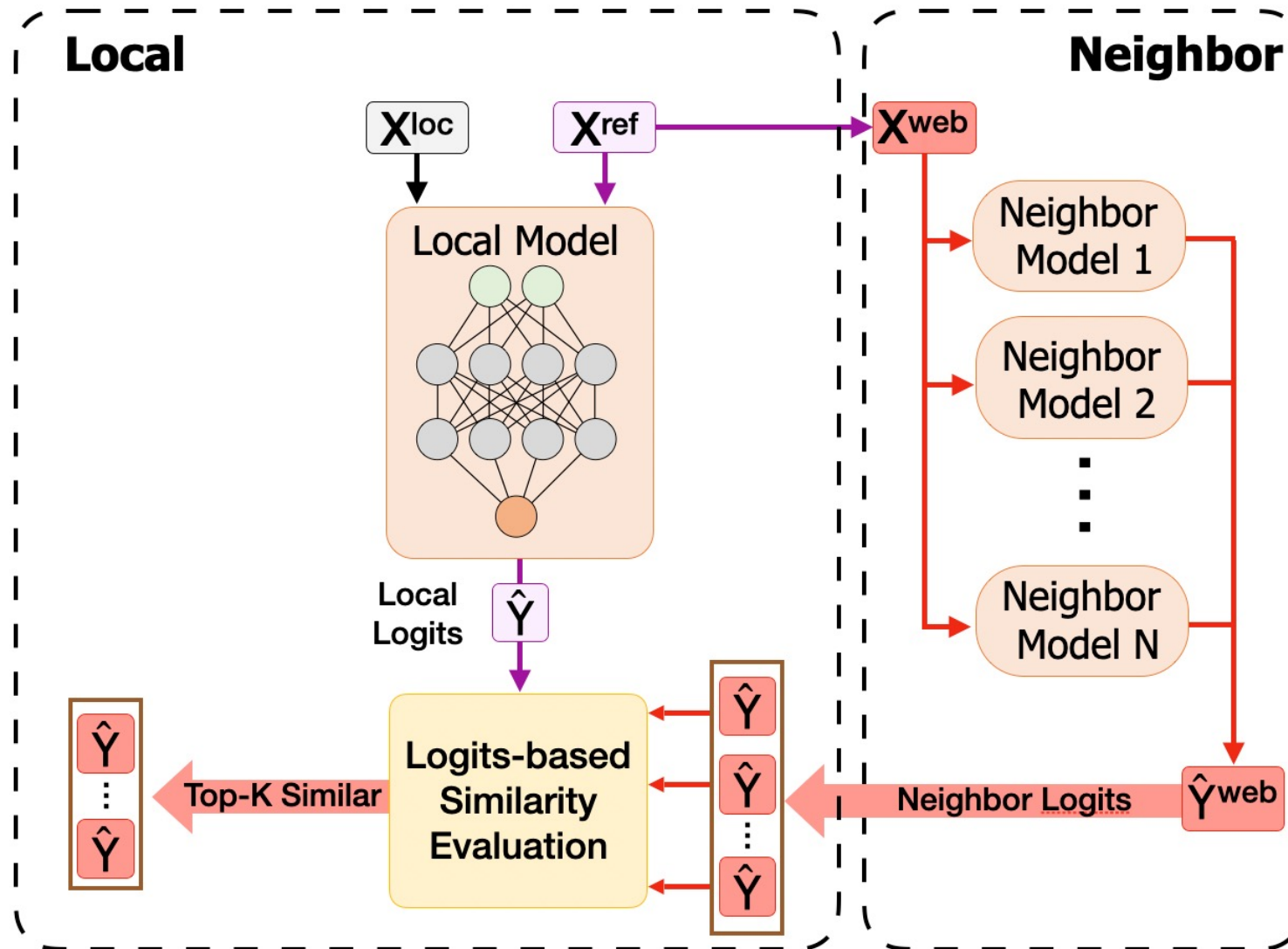


Figure 4(a). Logits-based P2P Similarity Evaluation.

Logits-based P2P Model Quality Assessment

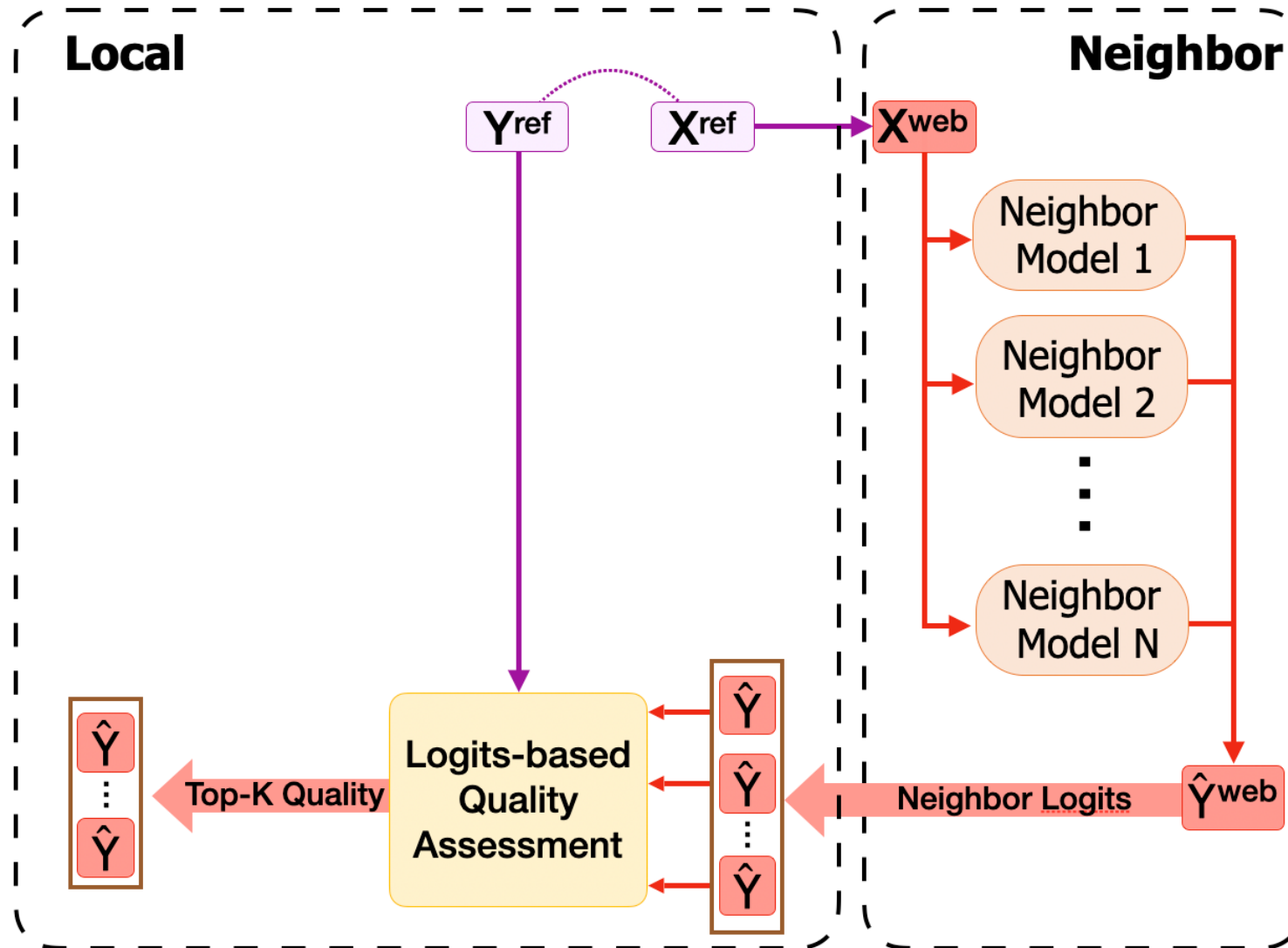


Figure 4(b). Logits-based P2P Model Quality Assessment.

Sounds interesting in FL neighbor selection.

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Can we make it scalable?

Workflow

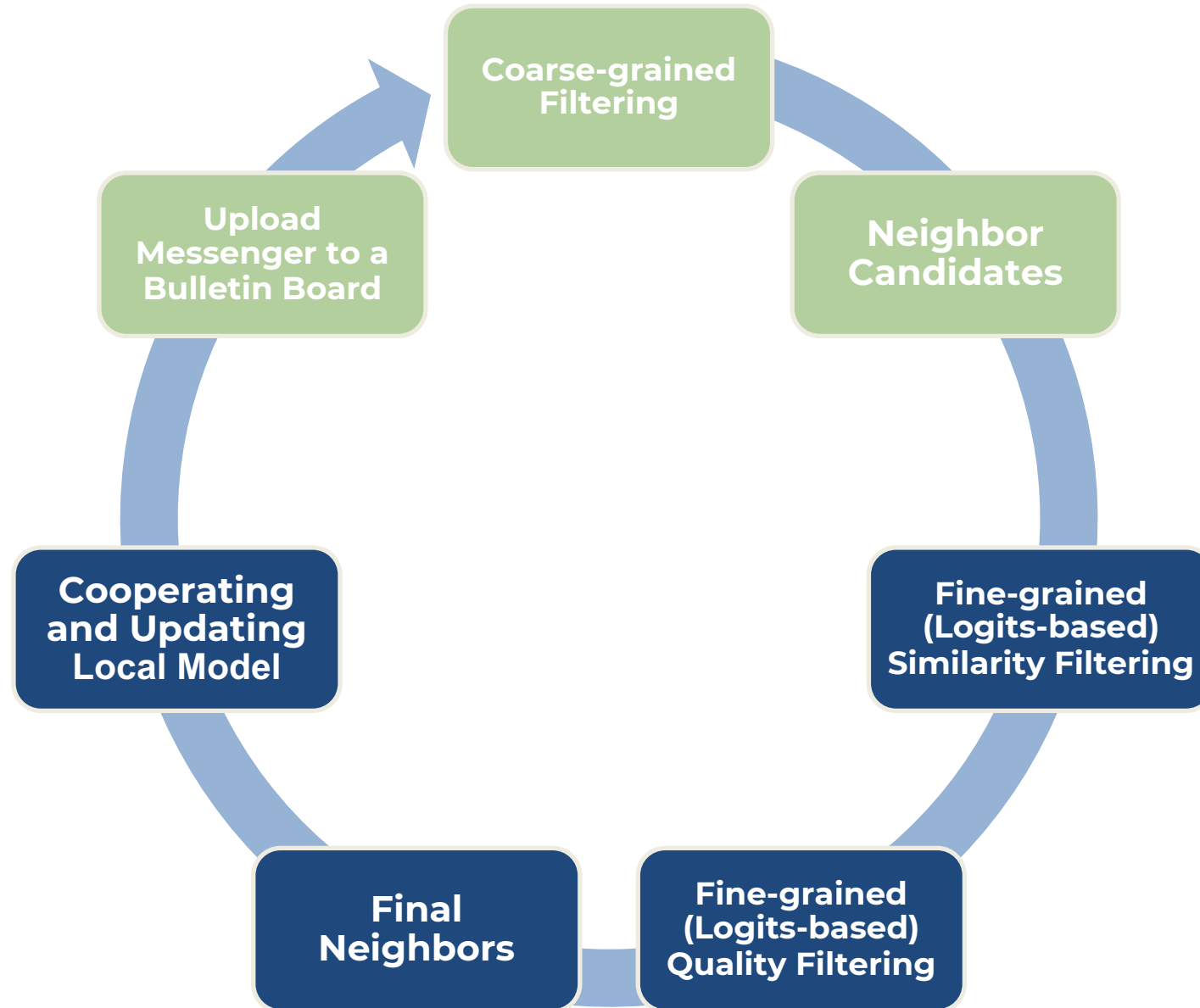


Figure 5. Workflow of Verifiable P2P Federated Learning.

Coarse-grained Similarity via Locality-Sensitive Hash (LSH)

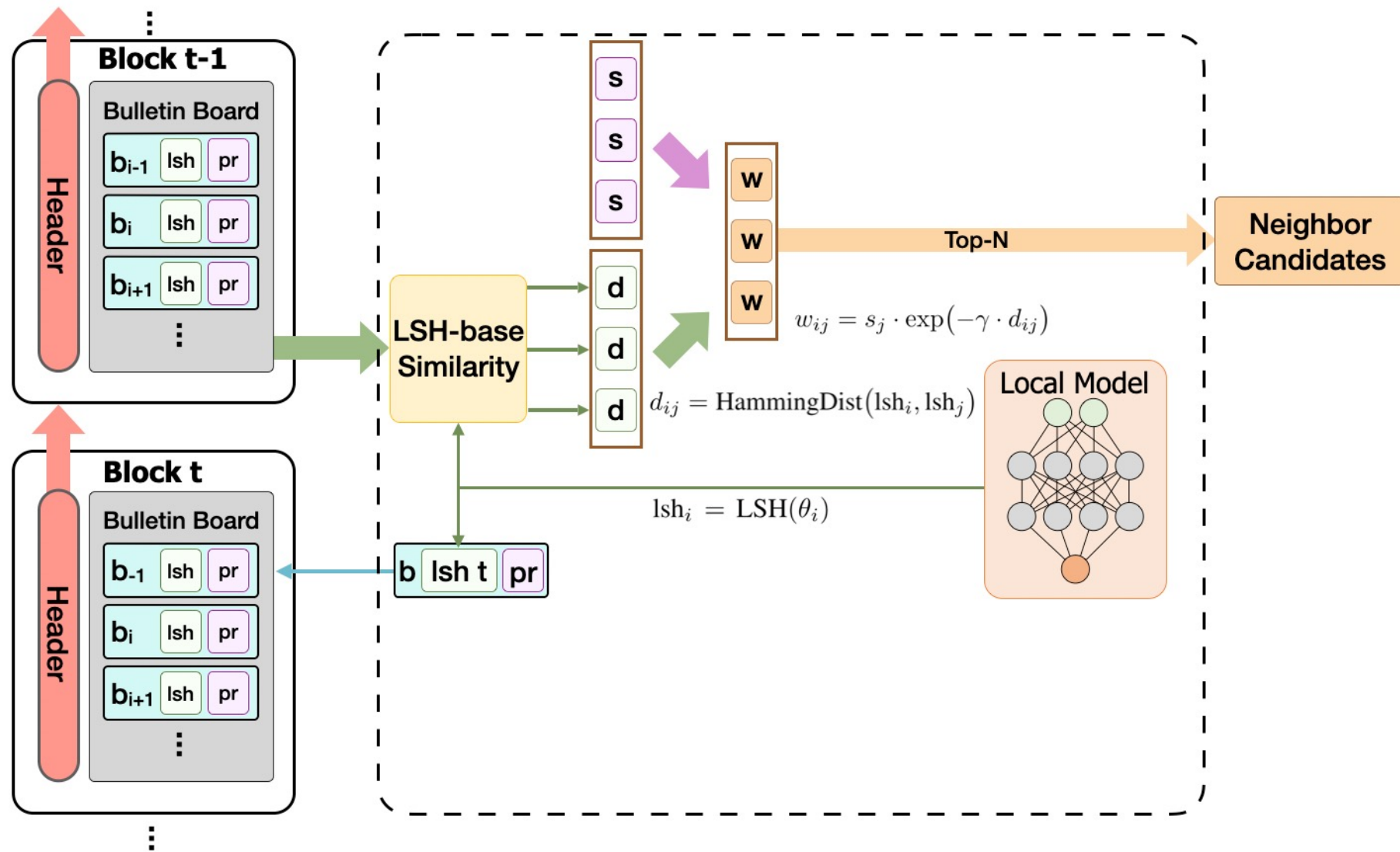


Figure 6(a). Coarse-grained Global Filtering - LSH.

Coarse-grained Model Quality via Peer Ranking (PR)

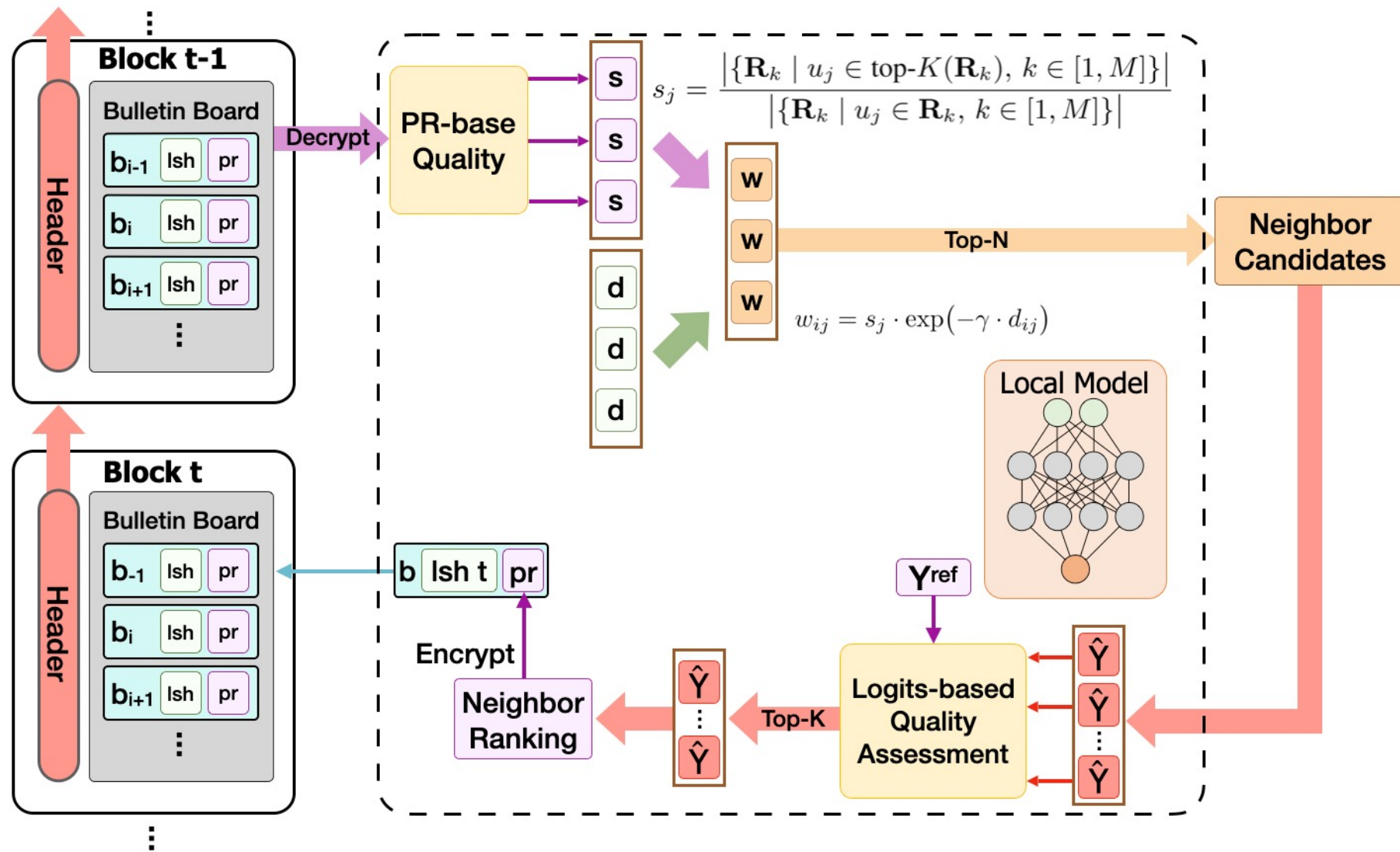


Figure 6(b). Coarse-grained Global Filtering - PR.

Coarse-grained Global Filtering with Bulletin Board

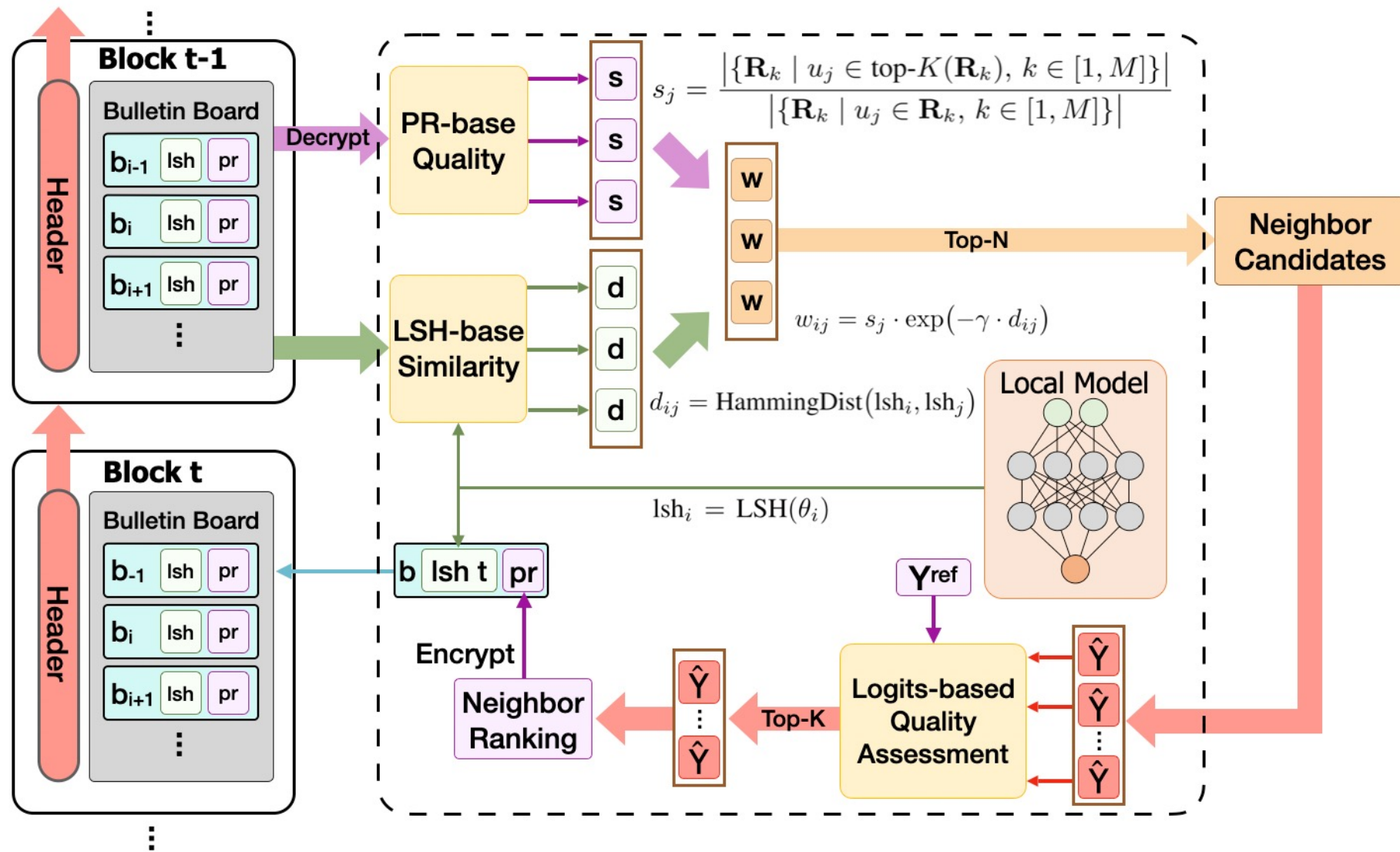


Figure 6(c). Coarse-grained Global Filtering with Bulletin Board.

Workflow

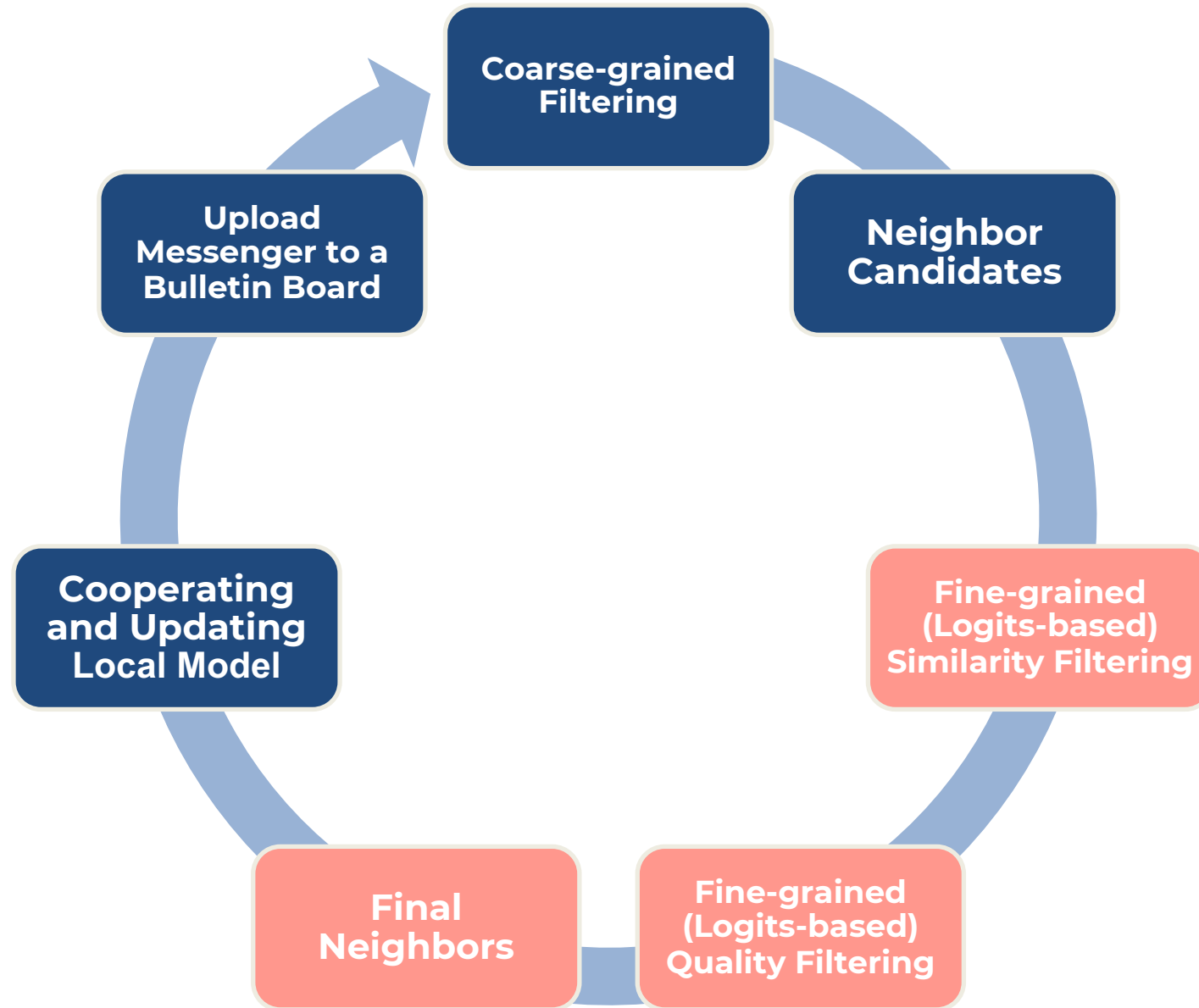


Figure 5. Workflow of Verifiable P2P Federated Learning.

Trust-free Personalized FL(TPFed) Architecture

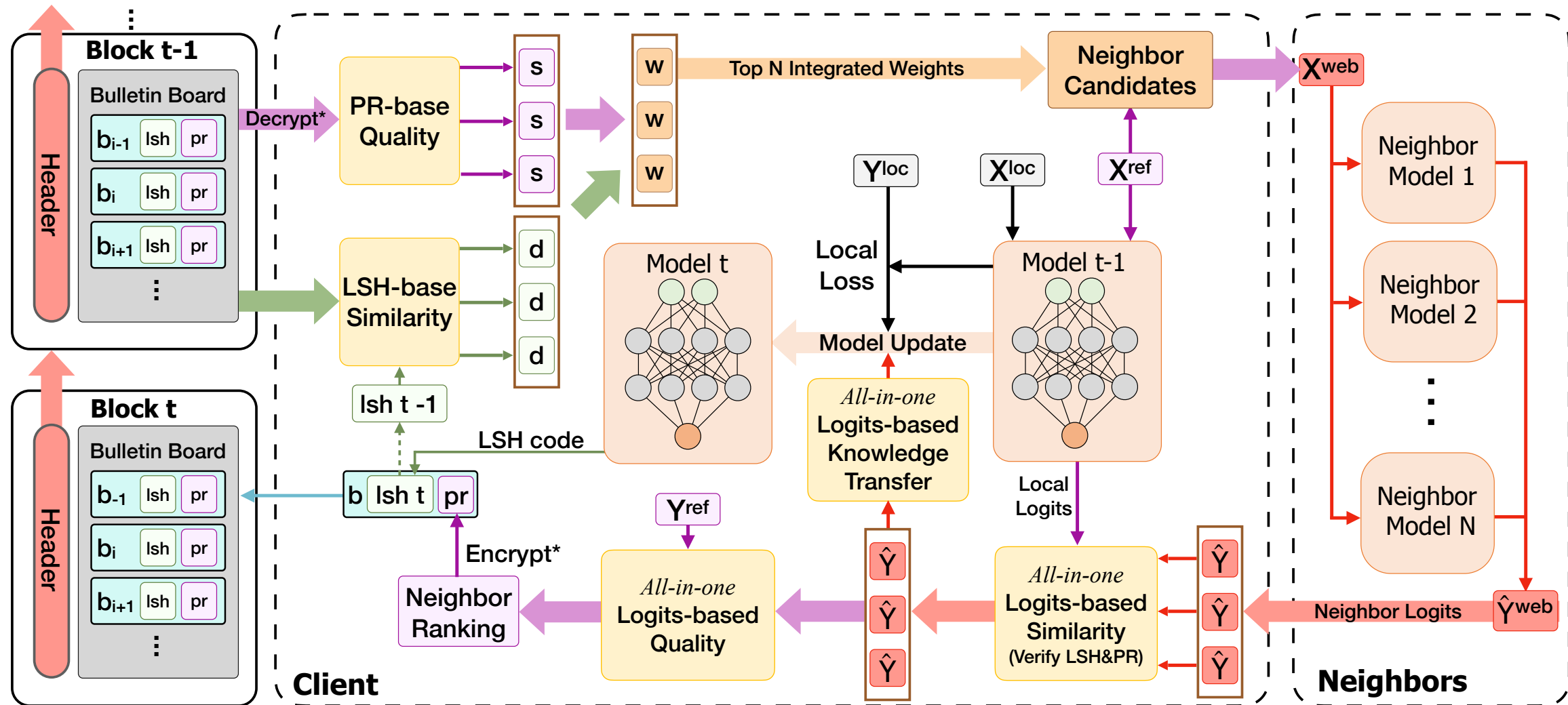


Figure 7. System architecture of Trust-free Personalized FL(TPFed).

New Challenge 1: LSH Forgery

LSH Forgey

Attack Method: Adversaries forge LSH codes to minimize Hamming distance (d_{ij}) artificially.

Deception: Tricking the victim into selecting malicious nodes as high-priority neighbors.

Impact: Vulnerable to DoS Attacks, where checking massive fake candidates exhausts bandwidth.

Solution

Strict Consistency Check: Repurpose the received Logits ($\hat{Y}_j^{\text{web}}$) for cross-verification.

Mechanism: Compare the Claimed LSH Distance against the Actual Semantic Divergence.

Defense Logic: If the deviation exceeds threshold τ , the check fails ($V_{ij} = 0$), proving the LSH contradicts actual behavior.

New Challenge 2: PR Collusion

PR Collusion

Attack Method: Malicious cliques mutually boost each other's Ranking Scores s_j .

Deficiency: Commit-and-reveal schemes prevent tampering but cannot detect pre-planned collusion.

Impact: Attackers gain falsely inflated global reputation to mislead honest clients.

Solution

Heuristic Strategy: Treat global ranking s_j merely as a discovery signal, not ground truth.

Local Dominance: Final weight is dominated by Local Logits Evaluation L_i^{ref} .

Defense Logic: High local loss suppresses the weight to zero w_{ij}^{final} to 0, rendering the inflated reputation useless.

More Details

Implementation Concerns & Solution

Blockchain Efficiency: Real-time on-chain updates incur high latency and gas costs.

Solution: Asynchronous Updates T_{sync} . Decouple reporting from training; publish LSH codes only every T_{sync} rounds.

LSH Validity: Neural network permutation invariance can invalidate parameter-based LSH.

Solution: Shared Initialization $\theta^{(0)}$. Aligns parameter spaces across clients, ensuring LSH effectiveness.

Cold Start Problem: Strict ranking ignores new or unranked peers ("rich-get-richer").

Solution: ϵ -Greedy Exploration. Randomly sample $\epsilon \cdot N$ neighbors to ensure network diversity and discovery.

DoS Attack. Malicious nodes spam requests or rejoin repeatedly.

Solution: Asymmetric Cost. Local Blacklists T_{ban} force attackers to pay Gas fees for new identities, making attacks economically unviable.

Overfitting: Adversaries may memorize static reference data to pass checks.

Solution: Randomized Querying. Apply random augmentations to queries, forcing peers to generalize rather than memorize.

TPFed Algorithm

Algorithm 1 TPFed Framework for Client u_i

Require: Local $\{\mathbf{X}_i^{\text{loc}}, \mathbf{Y}_i^{\text{loc}}\}$, Reference $\{\mathbf{X}_i^{\text{ref}}, \mathbf{Y}_i^{\text{ref}}\}$, Init θ_i , Iterations T , Hyperparams $(\alpha_i, \gamma, \epsilon)$, Neighbors N

- 1: **for** $t = 1$ **to** T **do**
- 2: **Neighbor selection**
- 3: **if** $t > 1$ **then**
- 4: Retrieve $\mathbf{b}_j^{(t-1)} = \{\text{lsh}_j^{(t-1)}, C_j^{(t-1)}\}$ and $\mathbf{R}_j^{(t-1)}$; verify $\mathbf{R}_j^{(t-1)}$
- 5: **end if**
- 6: For each u_j : compute d_{ij} (Eq. (2)) and s_j (Eq. (3))
- 7: $w_{ij} \leftarrow s_j \exp(-\gamma d_{ij})$; $\mathcal{N}_i^{(t)} \leftarrow \text{Top}-(1-\epsilon)N$ by $w_{ij} \cup \text{Random-}\epsilon N$
- 8: **Communication**
- 9: **for** $u_j \in \mathcal{N}_i^{(t)}$ **do**
- 10: Send $\mathbf{X}_i^{\text{ref}}$ to u_j , receive $f(\theta_j^{(t)}, \mathbf{X}_i^{\text{ref}})$
- 11: $\ell_{ij}^{(t)} \leftarrow \ell(f(\theta_j^{(t)}, \mathbf{X}_i^{\text{ref}}), \mathbf{Y}_i^{\text{ref}})$
- 12: **end for**
- 13: **Model update**
- 14: Calculate weights $p_{ij} \propto w_{ij}^{\text{final}}$ (Eq. 6) using $\ell_{ij}^{(t)}$ and consistency $\mathcal{V}_{ij}$
- 15: $\bar{f}^{(t)} \leftarrow \sum_{u_j \in \mathcal{N}_i^{(t)}} p_{ij} \cdot f(\theta_j^{(t)}, \mathbf{X}_i^{\text{ref}})$
- 16: $\mathcal{L}^{(t)} \leftarrow \alpha_i \ell(f(\theta, \mathbf{X}_i^{\text{loc}}), \mathbf{Y}_i^{\text{loc}}) + (1 - \alpha_i) \|f(\theta, \mathbf{X}_i^{\text{ref}}) - \bar{f}^{(t)}\|^2$
- 17: $\theta_i^{(t+1)} \leftarrow \text{Update}(\theta_i^{(t)}, \nabla_{\theta} \mathcal{L}^{(t)})$
- 18: **Bulletin publication**
- 19: $\text{lsh}_i^{(t)} \leftarrow \text{update via Eq. (4)}$; $\mathbf{R}_i^{(t)} \leftarrow \text{sort by ascending } \ell_{ij}^{(t)}$
- 20: $C_i^{(t)} \leftarrow \text{Hash}(\mathbf{R}_i^{(t)})$; publish $\mathbf{b}_i^{(t)} = \{\text{lsh}_i^{(t)}, C_i^{(t)}\}$
- 21: Reveal $\mathbf{R}_i^{(t)}$ for verification at $t+1$
- 22: **end for**

1. Introduction

2. Methodology

▶ 3. Experiments

4. Conclusion

Research Questions(RQs)

RQ1 (Effectiveness): Does TPFed outperform existing centralized, semi-decentralized, and decentralized federated learning baselines in terms of model accuracy?

RQ2 (Identity Defense): Can TPFed effectively detect and reject adversaries attempting to spoof model similarity (LSH forgery) to perform impersonation attacks?

RQ2 (Poisoning Resilience): How robust is TPFed against active poisoning attacks where malicious clients inject noise to disrupt the training process?

RQ4 (Scalability): How does the communication overhead of TPFed compare to other methods, and does the TPFed-idx variant effectively improve scalability?

RQ5 (Ablation): What is the contribution of each core component (LSH, Peer Ranking, and Logits Verification) to the overall framework performance?

RQ6 (System Robustness): Can TPFed maintain stability under adverse system conditions, such as network delays, transmission failures, or low-quality reference data?

RQ1: Performance Effectiveness

TABLE I
PERFORMANCE COMPARISON OF DIFFERENT FRAMEWORKS ACROSS BASE MODELS AND DATASETS

Category	Framework	P-MNIST	A-ECG	S-EEG	CIFAR-100
SILO	SILO	0.8774 ± 0.0088	0.9112 ± 0.0050	0.8367 ± 0.0090	0.4210 ± 0.0145
CFed	BlockFL	0.9342 ± 0.0011	0.9117 ± 0.0016	0.8358 ± 0.0045	0.4510 ± 0.0028
	FedMD	0.9375 ± 0.0017	0.9116 ± 0.0016	0.8324 ± 0.0050	0.4630 ± 0.0020
SFed	pFedClub	0.9395 ± 0.0019	0.9176 ± 0.0021	0.8421 ± 0.0077	0.4805 ± 0.0041
	SQMD	0.9379 ± 0.0032	0.9124 ± 0.0030	0.8446 ± 0.0063	0.4850 ± 0.0035
DFed(G)	ProxyFL	0.9224 ± 0.0051	0.9051 ± 0.0066	0.8391 ± 0.0091	0.4701 ± 0.0053
	KD-PDFL	0.9232 ± 0.0033	0.9106 ± 0.0050	0.8444 ± 0.0088	0.4623 ± 0.0048
DFed(W)	TPFed-idx	0.9375 ± 0.0023	0.9156 ± 0.0059	0.8384 ± 0.0060	0.4937 ± 0.0042
	TPFed	0.9422 ± 0.0041	0.9287 ± 0.0038	0.8525 ± 0.0065	0.5295 ± 0.0050

RQ2: Resilience to Identity Attacks

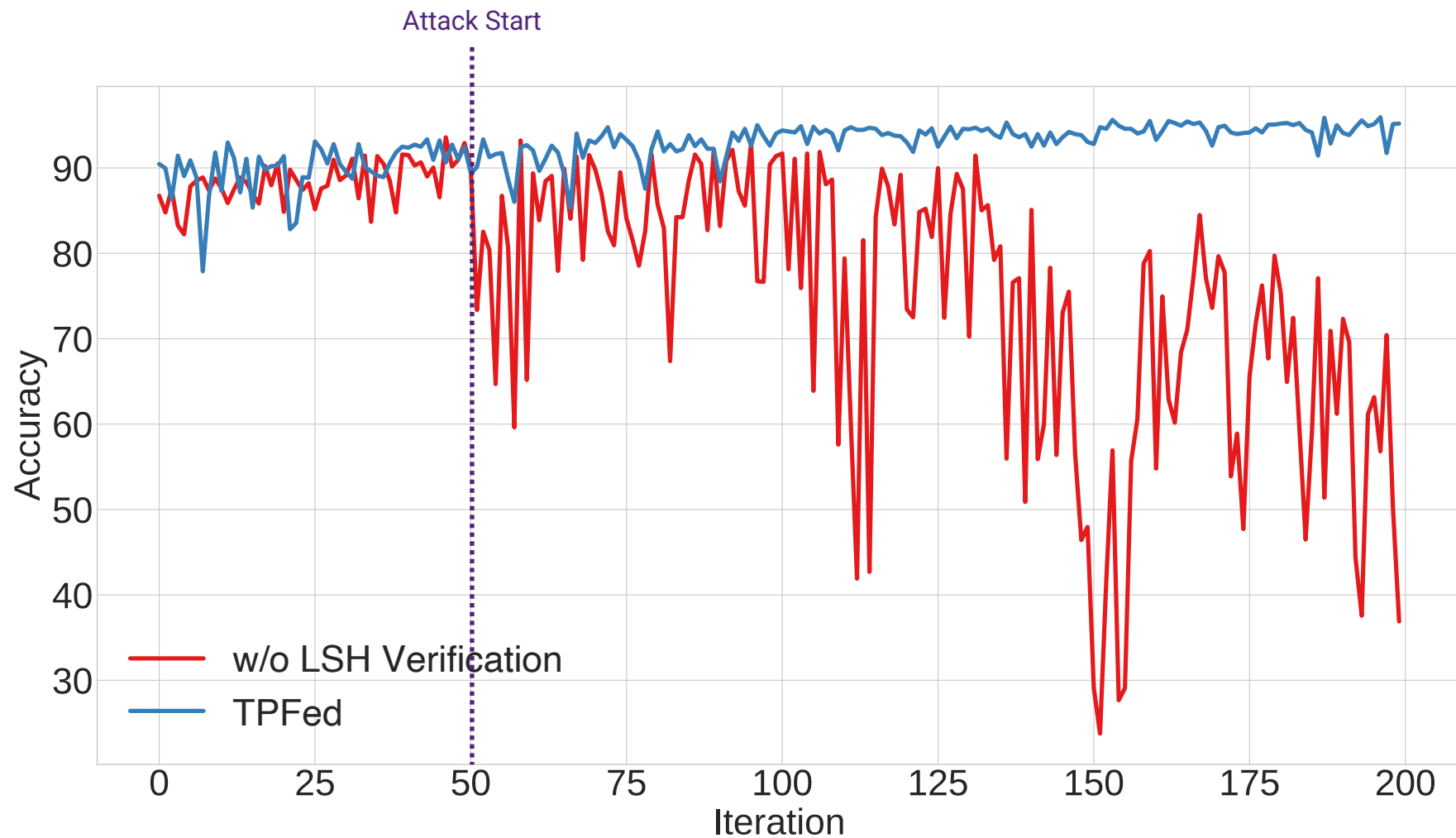


Figure 8. Accuracy stability under LSH Cheating Attack..

RQ3: Resilience to Poisoning

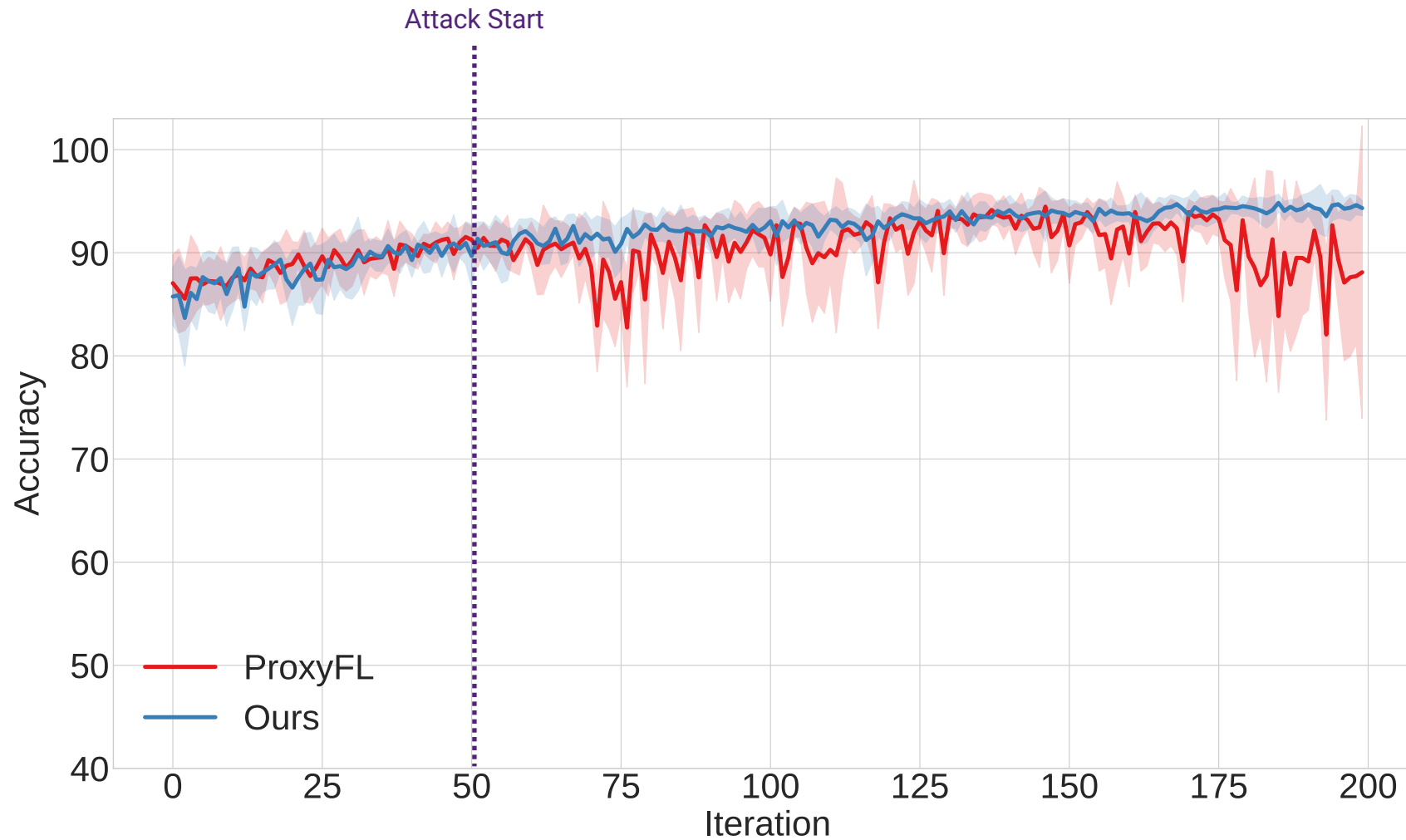


Figure 9(a). Poison attack with 20% clients act as adversaries.

RQ3: Resilience to Poisoning

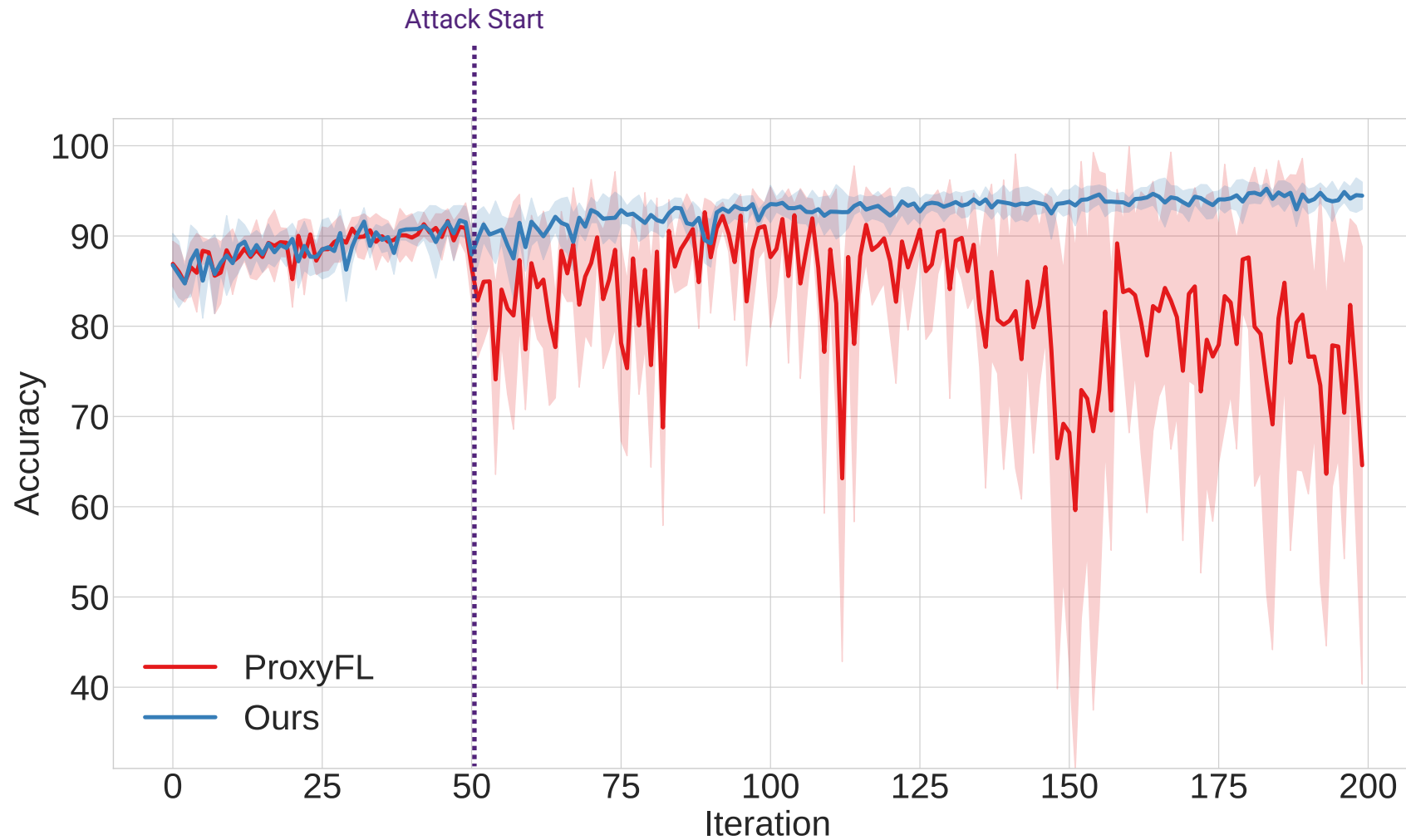


Figure 9(b). Poison attack with 40% clients act as adversaries.

RQ3: Resilience to Poisoning

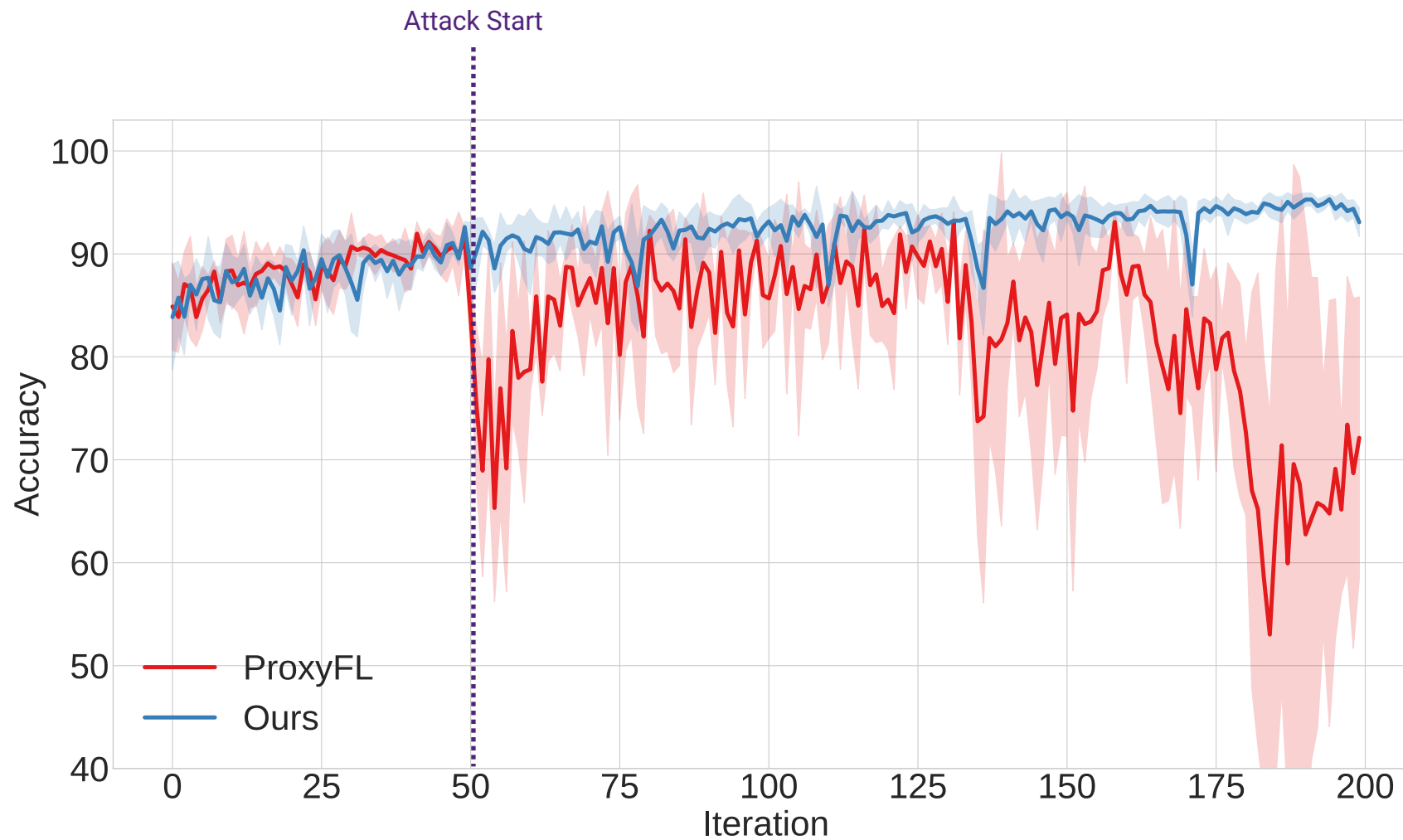


Figure 9(c). Poison attack with 60% clients act as adversaries.

RQ4: Scalability

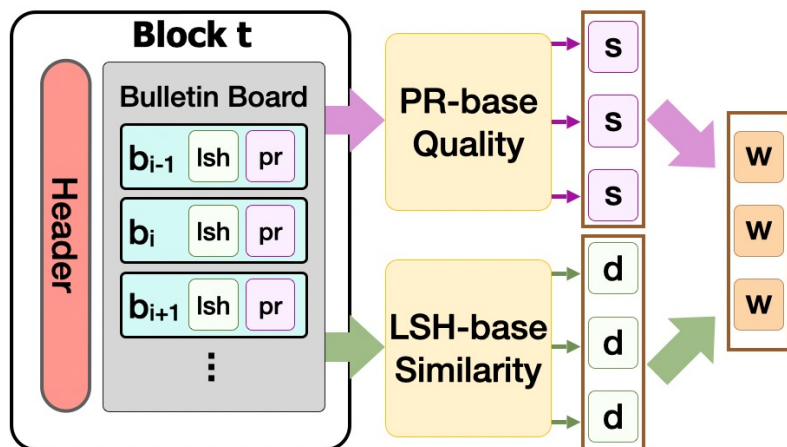


Figure 10(a). Coarse-grained Global Filtering of TPFed

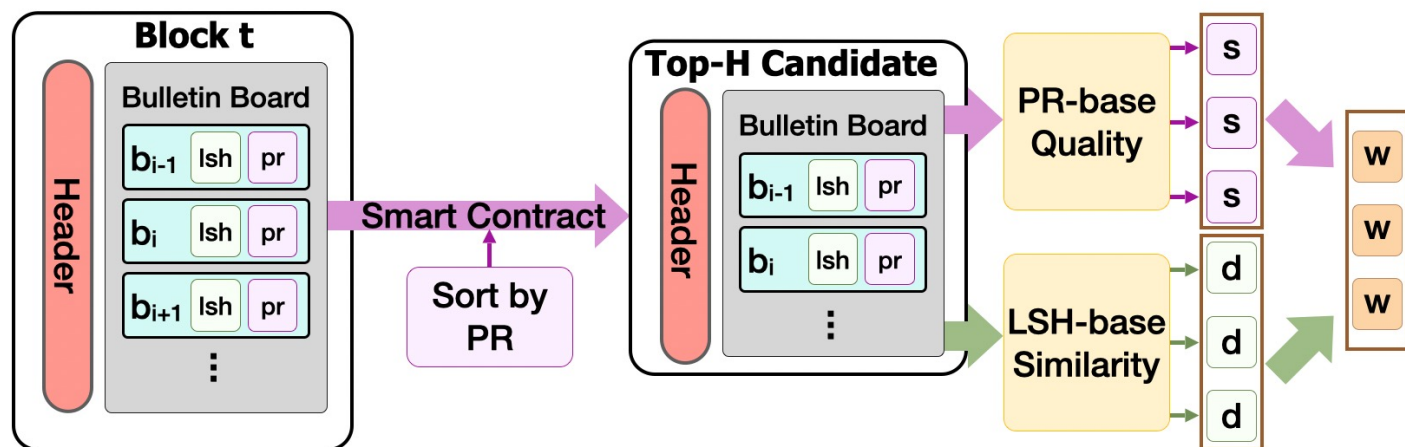


Figure 10(b). Coarse-grained Global Filtering of TPFed-idx

The Bottleneck of TPFed

Reason: Each client must parse and sort all M candidates from the network.

Complexity: $O(M \log M)$ per client.

Consequence: Computation overhead spikes as network size M grows.

TPFed-idx

Offloading: The Smart Contract performs a one-time global sort based on s_j .

Indexing: Clients only download the Top-H candidates $H \ll M$.

Efficiency: Complexity drops to $O(H \log H)$.

Trust: The index is verifiable, preserving decentralization.

RQ4: Scalability

TPFed-idx

Experimental Setup

- Network sizes $M = \{20, 100, 500\}$.

Observation 1 (Overhead)

- TPFed: Overhead grows significantly (1.17 $\rightarrow$ 1.68) as M increases.
- TPFed-idx: Overhead remains constant (1.17) regardless of network size.

Observation 2 (Accuracy)

- TPFed-idx maintains competitive accuracy (minor trade-off for scalability).

TABLE III
COMPARISON OF TPFED AND TPFED-IDX WITH $H=20$ ON CIFAR-100.
(M IS THE NUMBER OF CLIENTS. H IS THE RETRIEVE RANGE. OVERHEAD
PER ROUND IS NORMALIZED TO FEDMD)

M	TPFed Acc	TPFed Overhead	TPFed-idx Acc	TPFed-idx Overhead
20	0.4505	1.17	0.4505	1.17
100	0.5295	1.25	0.4937	1.17
500	0.5323	1.68	0.4991	1.17

RQ5: Ablation Study

Ablation

Observation

- Removing LSH or PR causes 1–3% accuracy drop
- Removing logits verification causes the largest degradation
- **Conclusion:** all components matter, and logits-based verification is the most critical.

TABLE IV
ABLATION STUDY FOR TPFED

Config	MNIST	A-ECG	S-EEG	CIFAR
TPFed	0.9422	0.9287	0.8525	0.5295
w/o LSH	-0.0104	-0.0070	-0.0153	-0.0194
w/o PR	-0.0116	-0.0061	-0.0198	-0.0308
w/o logits	-0.0282	-0.0195	-0.0202	-0.0371

RQ6: System Robustness 1

Latency/Congestion Robustness

Scenarios

- 20% Delay / 40% Delay / 20% Failure.

Mechanism

- If a bulletin is not on-chain by the deadline → Skip (No waiting).
- Prevents the entire system from hanging due to slow nodes.

Observation

- Minimal accuracy loss (e.g., MNIST drops only ~1.4% even with 20% failure).

TABLE V

IMPACT OF DELAYED AND FAILED UPLOADS ON TPFED'S ACCURACY.

Dataset	20% Delay	40% Delay	20% Fail
MNIST	-0.0085	-0.0132	-0.0143
A-ECG	-0.0063	-0.0091	-0.0120
S-EEG	-0.0049	-0.0078	-0.0101
CIFAR	-0.0024	-0.0042	-0.0065

RQ6: System Robustness 2

Reference datasets (X^{ref}) in Low Quality

Scenarios

- Replaced 20%, 40%, and 60% of reference data with local noise.

Mechanism

- LSH/Rank Verification: Bad reference data leads to bad Logits.
- Randomized Querying: Prevents overfitting to specific noisy samples.

Observation

- Up to 40% Corruption: Robust performance (Accuracy drop < 2%).
- At 60% Corruption: Moderate degradation, but training remains stable.

TABLE VI
PERFORMANCE IMPACT FROM BIASED OR LOW-QUALITY REFERENCE
DATA ACROSS DIFFERENT CORRUPTION RATES.

Dataset	20% Corr.	40% Corr.	60% Corr.
MNIST	-0.0066	-0.0104	-0.0189
A-ECG	-0.0011	-0.0023	-0.0055
S-EEG	-0.0005	-0.0013	-0.0037
CIFAR	-0.0008	-0.0015	-0.0024

1. Introduction

2. Methodology

3. Experiments

▶ 4. Conclusion

Contributions

Problem Identification:

- Identified the critical gap in trust-free verification (Quality & Similarity) for decentralized FL.

Framework Proposal (TPFed):

- A novel Web-based paradigm combining Blockchain coordination with LSH for global-scale neighbor discovery.

Core Mechanism:

- "All-in-one" P2P Distillation: Simultaneously achieves knowledge transfer, model quality evaluation, and similarity verification.

Empirical Success:

- Demonstrated superior Accuracy and Resilience against DoS and Poisoning attacks compared to SOTA baselines.

Limitations

Model Homogeneity Constraint:

- Issue: LSH relies on model parameters, requiring clients to share the same architecture and initialization.
- Mitigation: Redundancy mechanisms and Logits verification help mitigate false positives caused by permutation invariance.

Simple Incentive Mechanism:

- Issue: Current defense relies on Exclusion (Blacklisting).
- Gap: Lacks granular economic penalties or game-theoretical rewards for long-term sustainability.

Reference Data Dependence:

- Issue: Reference dataset are not available for all tasks.
- Context: Although our robustness tests show tolerance to noise (RQ6), this dependency restricts application in scenarios where no public data is available.

Future Work

Heterogeneous Model Support:

- Developing functional-based hashing or representation alignment techniques to support diverse model architectures.

Game-Theoretical Incentives:

- Integrating Tokenomics (Token Economics) or Reputation Credits to reward high-quality contributors and economically penalize malicious actors.

LLM MoE Collaboration:

- Each expert are specialized in certain sort of tokens, so the selective co-distillation (i.e., both similar & good expert are more valuable in collaboration, and we need a trust-free mechanism to find them).

Thank You